Mustafa Cavus, Ph.D.
Department of Statistics, Faculty of Science, Eskisehir Technical University, 26555 Eskisehir, Türkiye
Email: mustafacavus@eskisehir.edu.tr | **Date:** September 3, 2026

To:
Guest Editors, Special Issue on "Responsible and Trustworthy Artificial Intelligence in Tourism and Hospitality"
Editors-in-Chief, *Electronic Markets*, Springer Nature

Subject: Submission of Original Research Article for Special Issue Consideration

Dear Guest Editors and Editorial Board,

I am pleased to submit our original research manuscript entitled **"Explainable Hotel Booking Cancellation Prediction Models: Towards Trustworthy Survival Analysis in Hospitality"** for consideration for publication in *Electronic Markets* within the Special Issue on **"Responsible and Trustworthy Artificial Intelligence in Tourism and Hospitality"**.

Motivation and Alignment with the Special Issue:
As predictive algorithms increasingly govern hotel revenue management, reliance on opaque "black-box" systems creates acute vulnerabilities regarding operational stability, customer trust, and algorithmic accountability. Traditional hospitality literature frames cancellation forecasting as a static binary classification problem, which ignores *when* attrition events occur and introduces systematic predictive bias by treating active, right-censored bookings as non-canceled. To operationalize the core tenets of Responsible and Trustworthy AI, this study introduces an end-to-end explainable time-to-event (survival analysis) framework delivering three primary contributions:

1. **Methodological Advancement & Calibration:** We reframe cancellation forecasting as a continuous time-to-event problem, benchmarking 10 survival algorithms across two structurally distinct empirical cohorts ($N = 119{,}390$ and $N = 36{,}275$). We demonstrate that while tree ensembles maximize discriminative ranking (C-index), Oblique Random Survival Forests and Conditional Survival Forests achieve superior probability calibration (Brier score and Right-Censored Log-Loss), which is critical for revenue management.
2. **Continuous, Time-Dependent Explainability (XAI):** We introduce Survival Permutation Feature Importance (SurvPFI) and Partial Dependence Survival Profiles (SurvPDP) to hospitality. These methods track the evolving marginal impact of covariates across the booking lifecycle, empirically unmasking compounding price elasticity near check-in and detecting policy-driven structural anomalies.
3. **Actionable Decision Support & Responsible Governance:** We bridge algorithmic outputs with operational practice, demonstrating how auditable survival curves enable dynamic overbooking quotas, optimal re-marketing windows, and trustworthy algorithmic governance.

Author Declarations:

- This manuscript represents original work that has not been published previously and is not under consideration elsewhere.
- The author has approved the manuscript and agrees with its submission to *Electronic Markets*.
- The author declares no conflicts of interest or competing financial interests.
- To ensure transparency and reproducibility, all replication code and pipelines are publicly accessible via an anonymized open science repository: https://anonymous.4open.science/r/explainable-hotel-booking-cancellat

Thank you very much for your time, consideration, and handling of our manuscript.

Sincerely,

Mustafa Cavus, Ph.D.
Department of Statistics, Eskisehir Technical University
Email: mustafacavus@eskisehir.edu.tr